CRMhs Specification

A semantic model for Heritage Science

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Based on a preliminary and partial work by a wider team of researchers

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1. Introduction

Heritage Science is an interdisciplinary field devoted to the study, conservation, accessibility, and interpretation of cultural heritage. It integrates a broad spectrum of scientific activities in support of heritage materials, drawing upon methodologies and analytical frameworks from disciplines such as physics, chemistry, biology, and related sciences. It fosters the use of highly sophisticated analysis techniques and high precision instruments to avail a better reading of artefacts and artworks and a deeper understanding of their physical composition, dating, geographic and temporal provenance.

A scientific approach to the study of cultural objects is essential to address questions that cannot be resolved through historical or stylistic analysis alone. It may, for instance, contribute to:

- assessing the condition of artefacts in order to inform appropriate conservation and restoration strategies
- identifying materials and production techniques
- tracing the provenance of raw materials to reconstruct historical trade routes
- dating organic or inorganic matter
- supporting assessments of authenticity

Unlike domains traditionally associated with large scale data production, for instance nuclear physics, Heritage Science is typically characterised by the generation of numerous small and highly contextualised datasets. This situation arises from several converging factors. One of these is the long-standing practice of privileging the interpretative outcomes of scientific analyses over the preservation of the underlying numerical data, which are often regarded as instrumental and expendable once their immediate purpose has been fulfilled. As an example, pigment analyses performed on a painting may contribute decisively to its dating or attribution, while the quantitative measurements of elemental fluorescence obtained during the analysis are rarely retained or made accessible beyond the published interpretative conclusions. A further contributing factor is the pronounced heterogeneity of heritage science data, which reflects the wide diversity of materials, objects, analytical techniques, and research questions involved. This heterogeneity is compounded by the coexistence of multiple documentation standards and recording practices, each rooted in specific disciplinary traditions and methodological approaches. Finally, the absence of a mature and integrated ecosystem capable of demonstrating the tangible benefits of data driven research and systematic data reuse has historically limited the motivation of the researchers to invest in structured data management and dissemination. Taken together, these conditions have resulted in a research landscape where heritage science data remain largely fragmented, weakly documented, and difficult to integrate or reuse beyond their original analytical context.

Conversely, the integration of heterogeneous datasets across institutional and disciplinary boundaries would substantially expand opportunities for knowledge advancement. By enabling the combined analysis of data derived from different objects, contexts, and investigations, such integration would support a more comprehensive understanding of the

shared histories of cultural artefacts and facilitates more informed strategies for their conservation, management, and long-term preservation. The efficient management of scientific information and its coherent integration at the digital level therefore represent key conditions for fully exploiting the potential of heritage science research. Organising the entirety of data related to a given artefact and making it accessible to the research community can effectively reactivate information that would otherwise remain confined to individual publications. This, in turn, enables new analytical perspectives to be applied to existing datasets, addressing novel research questions without the need to repeat measurements that are often costly, invasive, or irreversible.

In this context, information science offers a range of advanced methods and technologies for integrating, accessing, and exploiting heterogeneous forms of knowledge. Semantic data structures, together with artificial intelligence and machine learning approaches, enable computational systems to identify, relate, and retrieve interconnected information in accordance with specific research themes and questions. They also support complex queries over integrated digital archives that would be impractical or extremely time consuming to perform manually. The ongoing evolution of the digital research landscape is thus increasingly driven by the adoption of shared standards, ontologies, and controlled vocabularies, as well as by the availability of data in machine interpretable formats such as Linked Open Data. This process of standardisation opens new perspectives for data sharing and interoperability that were largely unattainable until recently, contributing to the realisation of a distributed yet semantically interconnected global knowledge space.

2. Existing models

The international research landscape offers numerous conceptual models designed to describe information generated by scientific investigation. Some of these are general in scope, while others are developed for specific disciplinary communities. However, before CRMhs no conceptual model has been explicitly conceived for Heritage Science as a distinct domain, capable of capturing the specific relationship between scientific inquiry and Cultural Heritage and of representing the full complexity of their interactions.

A defining characteristic of Heritage Science lies in its integration of rigorous analytical methodologies typical of the natural sciences with interpretative and contextual approaches rooted in the humanities. For this reason, prior to developing the CRMhs model, a systematic examination of existing conceptual frameworks in both domains, Cultural Heritage and Analytical Sciences, was undertaken with the aim of drawing upon established best practices and combining their respective strengths. Among these, some of the most relevant examples are highlighted below.

Within the broader scientific domain, several general-purpose models support the semantic representation of research information. Among these, CERIF[1] provides a structured metadata model for describing research entities and their interrelations, including an extensive vocabulary of roles and types commonly used in research contexts. Similarly, OBOE[2] offers an ontological framework for modelling and representing knowledge related to scientific

[1] https://eurocris.org/services/main-features-cerif.
[2] https://bioportal.bioontology.org/ontologies/OBOE.

observations. Both models supply classes and properties that enable detailed semantic descriptions of scientific activities and results.
Other initiatives are more domain specific and have been developed by large scientific communities to systematise discipline-oriented metadata. Examples include NeXus[3], a common data format for neutron, X ray, and muon science; AVM[4], a metadata framework for astronomical images; and CIF[5], a format for crystallographic information. The latter is particularly noteworthy for its modular structure and for the rich set of complementary resources it provides, including comprehensive domain dictionaries that enable a high degree of terminological standardisation. Additional terminological resources have been developed in fields such as mineralogy[6], while NASA maintains a broad and widely adopted general purpose thesaurus[7].
The Cultural Heritage domain likewise offers significant conceptual and terminological resources. Among these, CIDOC CRM[8] stands out as the most widely recognised conceptual model for the semantic representation of cultural heritage information. Its ecosystem of extensions provides a flexible and extensible framework capable of accommodating a wide range of heritage domains, from archaeology and museology to epigraphy and bibliographic documentation. Of particular relevance to the present work is the CRMsci extension, specifically designed to represent scientific observation and measurement processes. It constitutes the most comprehensive attempt to date to integrate scientific and humanistic information within a coherent conceptual structure. CRMhs builds upon this conceptual foundation and aims to produce records fully compatible with the CIDOC CRM ecosystem, and in particular with CRMsci. This alignment is essential, also in light of many major European research infrastructures and initiatives which have adopted CIDOC CRM as the backbone of their digital infrastructures.

3. CRMhs general overview

CRMhs is an ontology developed to capture and represent the full range of scientific activities involved in Heritage Science. It results from the integration of several existing ontologies that already provide robust models for general aspects of research, and introduces new entities specifically designed to address the distinctive features of scientific investigation in the context of cultural heritage. The development of CRMhs has also been informed by practical insights and modelling efforts conducted within the ARIADNE[9], 4CH[10] and ARTEMIS[11] European initiatives, with a particular focus on the description of scientific workflows and the representation of structured research data.

CRMhs was developed with the aim of:

[3] https://www.nexusformat.org.
[4] https://www.astropix.org/page/avm.
[5] https://www.iucr.org/resources/cif.
[6] https://webmineral.com.
[7] https://www.sti.nasa.gov/nasa-thesaurus/.
[8] https://cidoc-crm.org.
[9] https://www.ariadne-research-infrastructure.eu.
[10] https://www.4ch-project.eu.
[11] https://www.artemis-twin.eu.

- providing a conceptual framework for institutions that need to organise, document, and preserve the results of scientific investigations, especially those that have not yet established a structured digital archive or laboratory repository;
- offering a semantic tool for the standardisation, sharing, and integration of scientific data across laboratories and institutions, enabling alignment with the wider cultural heritage domain and promoting interoperability between scientific and humanistic research environments.

To achieve these objectives, CRMhs adopts the data modelling principles of the CIDOC CRM and aligns with the FAIR principles12 to ensure the accessibility, integration, and reusability of digital scientific data. These principles have guided the development of the model both in its conceptual foundation and in the formal structuring of its classes and properties.
CRMhs is composed of classes and properties specifically defined to express the essential entities and relations that occur within Heritage Science investigations. These elements reflect a controlled and semantically coherent structure that avoids unnecessary complexity while ensuring completeness and usability across disciplines. The model is designed to be modular and extensible, allowing further specialisation or alignment with additional ontologies when required. This makes it adaptable to a variety of research infrastructures and documentation systems. Its primary purpose is to provide a structured representation of scientific investigations applied to cultural heritage materials, including the preparation, analysis, interpretation, and digital documentation phases. This structured approach facilitates data traceability, reproducibility, and integration within larger knowledge ecosystems.

CRMhs adopts the CIDOC CRM ontology as its conceptual foundation and ensures full semantic interoperability through explicit mappings to CRM entities and properties. The model is also aligned with international data standards relevant to the cultural heritage domain.

Table 1 below lists the ontologies and reference models that have been adopted or extended in CRMhs, including those used for scientific reasoning and digital provenance.

Model	Version	Prefix	Description	Classes prefix	Properti es prefix
CIDOC CRM	7.1.3	crm	A formal ontology for modelling Cultural Heritage information	E	P
CRMsci	2.0	crmsci	The scientific observation model	S	O
CRMdig	3.2.1	crmdig	Model for provenance metadata	D	L
CRMinf	1.2	crminf	An Extension of CIDOC-CRM to support argumentation	I	J

Table 1. Ontological models used to build the CRMhs model.

[12] https://www.go-fair.org/fair-principles.

From the conceptual point of view, Heritage Science consists of a series of activities aimed at enhancing historical and material knowledge about cultural objects, such as archaeological finds, artworks, and architectural elements, through scientific analysis applied directly to the objects or to selected portions or samples thereof. Scientific analyses and measurements are central to this discipline and can be described as sequences of coordinated activities, conducted according to rigorous protocols and criteria. Once codified and interpreted by experts, the results of these procedures lead to the acquisition of new empirical knowledge. These activities may be promoted or supported by various public or private initiatives, each with specific goals, within which scientific investigations are planned and performed.

The tools and techniques employed in Heritage Science are diverse and often differ significantly in their methodology and implementation. Many of these techniques are articulated into structured sequences of interrelated actions, such as the preparation of samples, the configuration and use of specific devices, and the processing and interpretation of data. In CRMhs, these phases are modelled as distinct yet connected scientific activities, allowing for modular representations of the overall workflow. The preparation of samples, for instance, may involve specific materials and treatments; the use of analytical devices requires setup and calibration; the processing of raw data demands appropriate software tools and expert interpretation. Notably, these activities may take place not only within laboratories but also directly *in situ*, such as in museums, private collections, or archaeological sites, particularly in cases where the objects are non-transportable due to size, fragility, or environmental constraints.

The procedures followed during these investigations are usually defined through detailed research protocols that ensure the scientific validity, effectiveness, and repeatability of the analyses. It is therefore crucial to document each activity accurately, capturing its purpose, conditions, and sequencing, in order to preserve the traceability and reliability of the results. Scientific results are typically recorded digitally, either through manual annotation or by automated systems embedded in the devices. The resulting files may take many forms, e.g., spreadsheets, databases, image files, numerical matrices, textual reports, and are often archived according to local laboratory standards. CRMhs provides a conceptual structure to describe both the files themselves and the full process that led to their creation. It also supports the design of interoperable digital platforms for the effective storage, retrieval, sharing, and reuse of such structured research data. Equally fundamental are the study objects themselves. Understanding their provenance, production methods, composition, stylistic and typological traits, geographical and chronological context, and other aspects commonly studied in the cultural heritage domain is essential for interpreting scientific data meaningfully.

Objects may be studied using different techniques depending on the analytical goals: some methods, such as XRF, are non-invasive and allow for the analysis of entire surfaces or selected areas without damaging the object. Other methods, such as radiocarbon dating, require the extraction of samples that are often consumed or rendered unusable in the process. CRMhs provides specific classes to model whole objects, defined areas, bounded portions, and extracted samples, enabling a fine-grained representation of how material entities are involved in scientific workflows. In cases involving fragmentary or incomplete objects, it is often necessary to document the original artefact of which the studied item is a

part. This is particularly important in the study of archaeological or biological remains, where information about the excavation context, preservation history, storage conditions and conservation treatments is critical to the reliability of the analysis. CRMhs supports such contextual documentation, enabling the correlation of scientific results with material, spatial, and historical provenance.

Study objects, their parts, and the samples derived from them represent the essential bridge between scientific investigation and humanistic inquiry. Their modelling requires the adoption of standardised data structures that allow for the integration and mutual intelligibility of information produced across disciplinary boundaries. Scientific investigations usually involve multiple actors and institutions, whose contributions must be explicitly recognised. These include public and private research bodies, universities, laboratories, and cultural institutions, as well as individual scientists and technical staff responsible for the preparation, calibration, analysis operation, and interpretation phases of the workflow. In CRMhs, the documentation of scientific activities is designed to interoperate with the CIDOC CRM core ontology, which already includes classes for objects, actors, institutions, and roles. Additional participants, such as museums, private collectors, archaeologists, curators, and other scholars, are often involved as holders, discoverers, or interpreters of the objects under study. Their roles, while not directly technical, are no less essential for framing the scientific process within a broader cultural and institutional context and must be documented accordingly.

4. Entities of the Heritage Science domain

This section presents an overview of the main entities introduced in CRMhs to model scientific activities and their context in the domain of Heritage Science. The conceptualisation builds on the principles outlined in the previous sections, aiming to capture the complexity of procedures, materials, devices, actors, and outputs involved in scientific investigations applied to cultural heritage.

As already stated, CRMhs adopts the terminological framework of CIDOC CRM (version 7.1.3) and its extensions, using the notions of class, subclass, property, domain, range, inversion, and inheritance to define the ontological structure. These notions underpin the semantic coherence and extensibility of the model, ensuring compatibility with international standards.

The ontology is organised around a hierarchy of entities, ranging from high-level concepts to specialised classes, tailored to the scientific processes of heritage investigation. Relationships between entities are defined through a set of properties, supporting the documentation of workflows, provenance, analytical procedures and data production.

One of the distinctive features and strengths of CRMhs lies in its hybrid structure, which combines domain-specific classes developed to capture the peculiarities of Heritage Science, with direct reuse of general-purpose classes and properties from the CIDOC CRM ecosystem. In particular, the entities that pertain to common aspects of cultural heritage documentation, such as time, place, actor and physical object, are modelled entirely through the existing semantic framework of CIDOC CRM, without redefinition or duplication. This approach ensures conceptual consistency, semantic interoperability, and seamless integration with other CRM-based ontologies.

Therefore, CRMhs reuses classes such as `E39 Actor`, `E53 Place`, `E52 Time-Span`, or `E77 Persistent Item`, focusing its modelling efforts on the scientific dimension to provide a set of specialised classes and properties representing scientific activities, analysis workflows, datasets, methods, devices and parameters. At the same time, all scientific processes can be linked directly to cultural heritage entities already defined in CIDOC CRM, such as `E22 Human-Made Object`, `E19 Physical Object`, or `E24 Physical Human-Made Thing`, thereby supporting a unified conceptual space where scientific and humanistic knowledge can be expressed together. This layered approach maximises reusability, ensures alignment with existing data standards, thus facilitating data sharing and integration across disciplinary boundaries.

The schema in

Figure 1 presents an overview of the classes and subclasses defined within the CRMhs model, providing a structured representation of its conceptual hierarchy. The diagram in

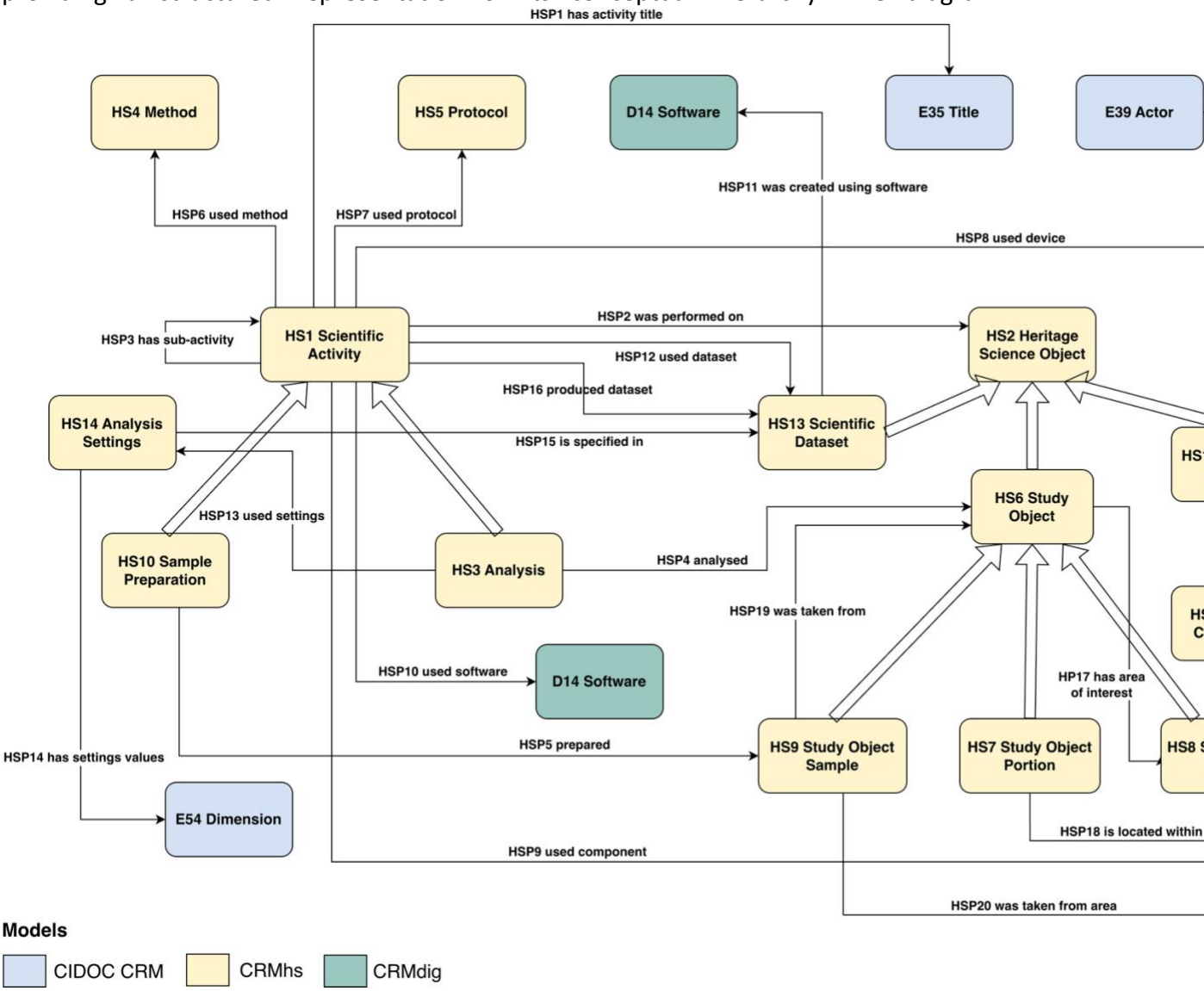

Figure *2* illustrates the properties defined within the CRMhs model, highlighting their scope and interconnections.

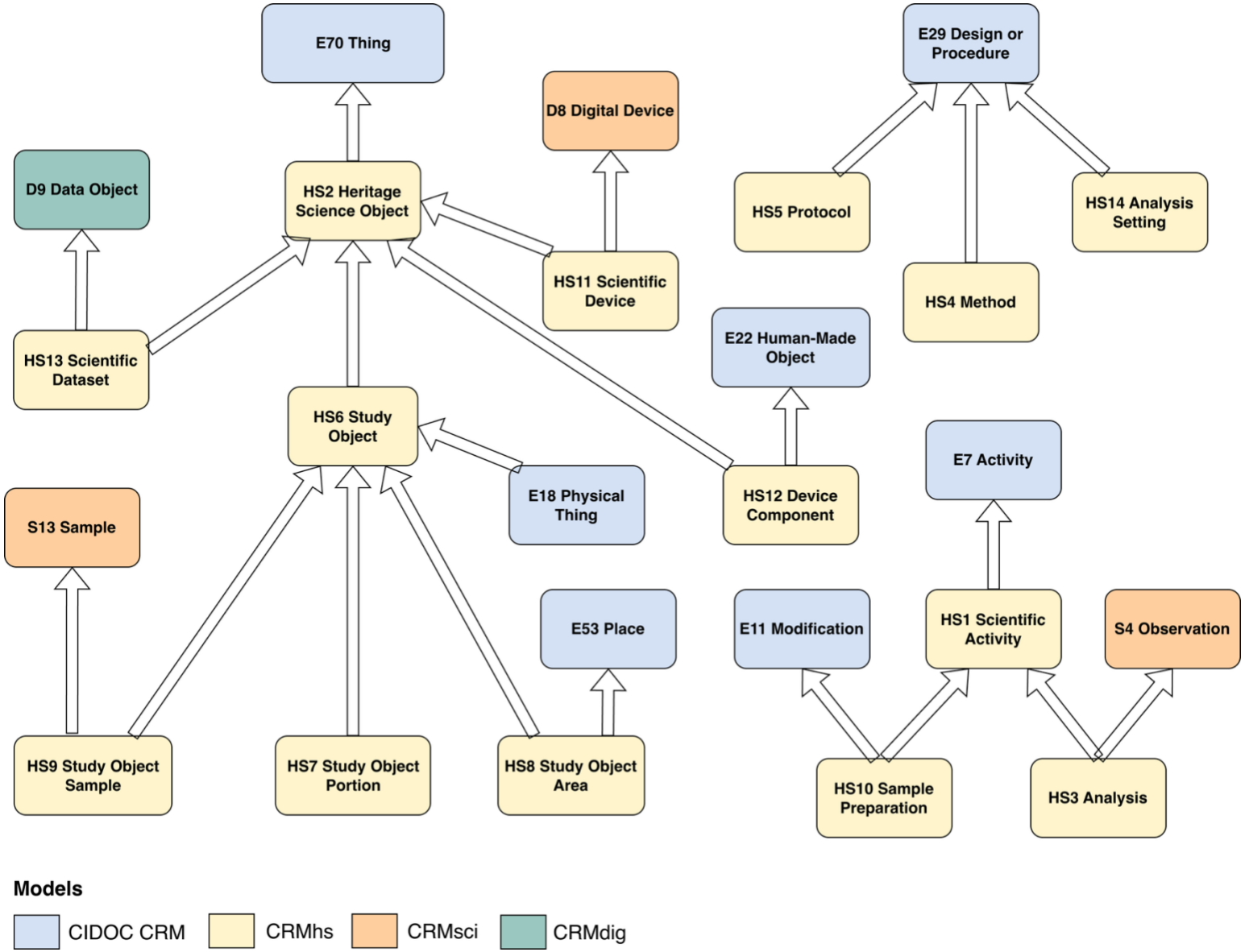

Figure 1. Hierarchy of the classes in CRMhs.

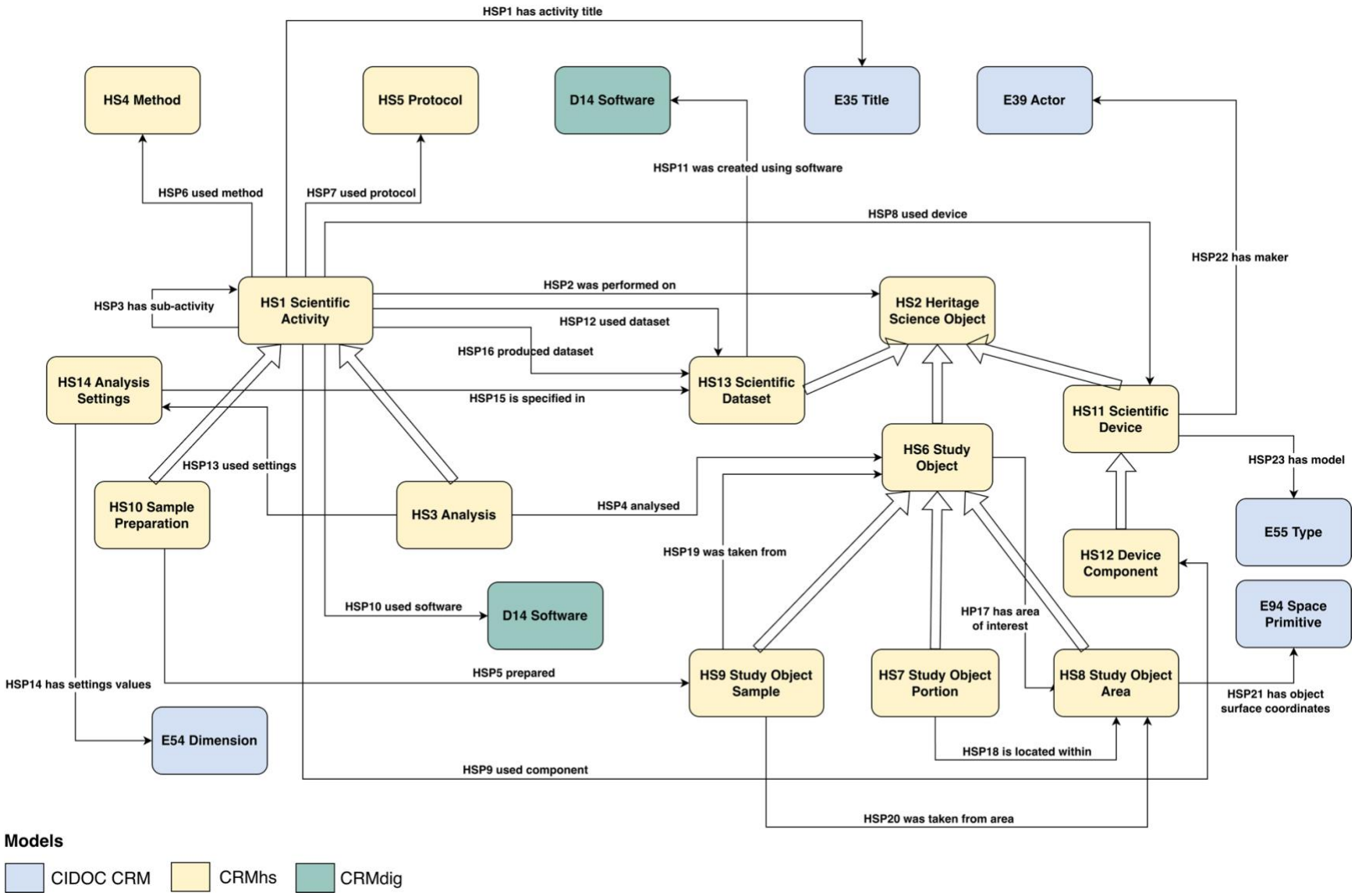

Figure 2. Classes and properties of the CRMhs model.

4.1. Events and activities of the Heritage Science research

Scientific work in the field of Heritage Science revolves around a wide range of coordinated activities, some experimental, some preparatory, others interpretative or supportive. These actions are often performed in specific temporal and spatial contexts, following rigorous standards and methodologies. CRMhs models these processes through a coherent set of entities designed to describe not only the analytical procedures themselves but also the broader operational framework that enables them.

4.1.1. Scientific activities

To represent the execution of scientific procedures, CRMhs introduces the top-level class `HS1 Scientific Activity`, designed to represent any kind of operation related to scientific investigation. These include the handling of objects, the preparation of samples, the operation of devices, the use of software, the collection and processing of data, and even the calibration and setup of tools.

Each scientific activity (`HS1`) is embedded in a broader procedural and contextual framework that captures not only what was done, but how, with what, and under which conditions. Activities can be decomposed into sub-activities using the property `HSP3 has sub-activity`, allowing the representation of complex workflows structured in discrete, traceable steps. The execution of an activity is often guided by formalised procedures. The property `HSP7 used protocol` links an activity to the protocol that governed its execution, while `HSP6 used method` connects it to a more general methodological framework. These elements provide a reference for reproducibility and comparability across investigations.

CRMhs also allows for the detailed documentation of the technical configuration of each activity. The properties `HSP8 used device` and `HSP9 used component` associate the activity with the devices and their components, while `HSP10 used software` specifies the digital tools involved. Finally, CRMhs supports the integration of pre-existing data into the scientific workflow through `HSP12 used dataset`, and the generation of new results via `HSP16 produced dataset`.

In keeping with the event-centric modelling approach of CIDOC CRM, each scientific activity in CRMhs can be temporally and spatially contextualised by means of the standard CRM classes `E52 Time-Span` and `E53 Place`, linked using the properties `P4 has time-span` and `P7 took place at`. This ensures full interoperability with existing heritage records and allows scientific procedures to be properly situated in both time and space.

4.1.2. Analytical procedures

The class `HS3 Analysis` is a specific subclass of `HS1 Scientific Activity` that captures those activities aimed at obtaining data through observation or measurement. This includes both destructive and non-destructive techniques and encompasses operations conducted on entire objects, specific portions, or isolated samples. Analyses are documented by specifically linking them to their object of study (`HSP4 analysed`). To account for the conditions under

which an analysis is made, the `HSP13 used settings` property is used to link it to one or more `HS14 Analysis Settings` entities, each of which may carry a value (`HSP14 has settings values`) or be specified in an external dataset (`HSP15 is specified in`). Additionally, the properties inherited from `HS1 Scientific Activity` enable the detailed documentation of analytical procedures, including the methods and protocols followed, the device and software used, the settings applied, and the resulting output.

4.1.3. Samples preparation

In many cases, analysis is preceded by specific preparatory procedures aimed at isolating or conditioning the sample. These operations are represented by the class `HS10 Sample Preparation`, another subclass of `HS1 Scientific Activity`, which includes cleaning, mounting, embedding, or other treatments required to make the sample analysable. Sample preparation is modelled in relation to the sample it transformed through the property `HSP5 prepared`, and it may itself produce new samples or involve components of the scientific device.

4.2. Physical and digital scientific objects

The domain of Heritage Science is characterised by the interplay between tangible materials and their digital representations. Scientific procedures involve the physical handling of cultural artefacts, the extraction and transformation of material samples, and the configuration of devices. At the same time, they produce and rely on digital outputs such as datasets, software, configuration files, and derived interpretations. CRMhs provides a comprehensive ontological framework to describe both dimensions, physical and digital, and to model their relationships within the research workflow.

4.2.1. Physical objects

Scientific analysis in the heritage domain is grounded in the materiality of the objects under investigation. These include not only the artefacts themselves, but also the physical environments, devices, and sampled substances that play a role in the analytical process. CRMhs captures this complexity through a set of dedicated classes, including the high-level class `HS2 Heritage Science Object` that represents all physical and digital entities involved in scientific investigations, while fully relying on the CIDOC CRM for documenting the cultural and historical aspects of these objects. Together, these physical entities constitute the core material dimension of Heritage Science, and their rigorous documentation, both scientific and cultural, is a foundational principle of CRMhs.

4.2.1.1. Study objects

The class `HS6 Study Object` represents the primary entities selected as the focus of scientific investigation. Conceptually defined by their epistemic role, these entities are not merely present within a scientific workflow but are actively examined through observation, measurement, or experimentation.

By being declared as a superclass of E18 Physical Thing, HS6 explicitly situates the scientific act of inquiry within the material branch of the CIDOC CRM ontology. Thus, HS6 is defined as a role-based aggregation analogous to E72 Legal Object, wherein diverse entities are brought together based on their status as subjects of rights. This modelling choice ensures that any physical entity, whether naturally occurring or human-made, may become a study object without altering its ontological status in the broader CRM framework. Instead, HS6 overlays an additional semantic layer that reflects intentional scientific engagement.

As a consequence, CRMhs does not redefine material entities already described in CIDOC CRM but reframes them according to their investigative role. All information concerning identity, provenance, typology, production, custody, and historical context continue to be expressed using CIDOC CRM classes and properties such as E22 Human-Made Object, E84 Information Carrier, P50 has current keeper, P102 has title, P108 has produced, and P12 occurred in the presence of.

This architectural alignment guarantees that scientific documentation extends rather than duplicates curatorial knowledge. Analytical results, datasets, and interpretative outcomes can therefore be directly connected to established heritage records and collection databases, preserving semantic continuity and ensuring interoperability within the CIDOC CRM ecosystem.

4.2.1.2. Heritage Science Samples

The class HS9 Study Object Sample, a direct subclass of HS6 Study Object and S13 Sample of CRMsci, represents discrete physical portions extracted from a study object (HS6) for the purpose of scientific investigation within a heritage context. Unlike a study object portion (HS7) that remains an integral part of the original object's physical fabric, an instance of HS9 is a newly individuated physical entity resulting from an act of separation. As a specialized heritage sample, it is characterized by a constitutive and enduring relationship with the object of provenance, distinguishing it from generic scientific specimens through its integration into the cultural entity's historical narrative. Therefore, a study object sample is not merely detached material but remains semantically and historically linked to the heritage object, often inheriting aspects of its identity, significance, and custodial responsibility. This continuing relationship implies that the sample is managed and preserved within a conservation framework, even when it undergoes transformation through sample preparations (HS10) or is partially consumed during analysis.

Through the property HSP19 was taken from, the model ensures a rigorous material traceability chain, preserving structured information regarding extraction conditions and the specific areas of origin (HS8). This identification and tracking of samples are essential for ensuring accountability and continuity of documentation, allowing scientific data to remain fully interoperable with curatorial and archaeological records.

4.2.1.3. Object portions, areas and localisations

In contrast to sampling, many Heritage Science investigations analyse cultural objects as a whole or focus on specific parts of it, either well-defined portions of matter or surface areas of interest. To represent this situation with precision, CRMhs introduces a set of dedicated

classes and properties. The class `HS7 Study Object Portion` models physically identifiable parts of a cultural object, such as a structural component, or a stratigraphic layer, that are the focus of specific scientific operations. These portions are often located within broader regions on the object's surface, represented by instances of the class `HS8 Study Object Area`, which is also defined as a subclass of the CIDOC CRM `E53 Place`. These areas serve as spatial anchors for scientific observations but also for tracking the provenance of samples.

To connect portions to the areas in which they are situated, the property `HSP18 is located within` is used. The provenance of a sample from a specific area of the object is instead recorded using the property `HSP20 was taken from area`, which allows the explicit modelling of this relationship, complementing the more general link to the parent object (`HSP19 was taken from`). Furthermore, the property `HSP17 has area of interest` links a `HS6 Study Object` directly to one or more `HS8 Study Object Area` instances. This enables clear documentation of which region(s) on an object were targeted by a particular investigation, even when no physical portion or sample is removed. Additionally, each `HS8 Study Object Area` can be precisely located on the object's surface using the property `HSP21 has object surface coordinates`, whose range is `E94 Space Primitive` from CIDOC CRM. This allows for the use of image-based references, relative positions, or any other spatial system appropriate to the context. Together, these elements ensure that the localisation of analyses and interventions is rigorously documented and reproducible.

4.2.1.4. Scientific devices and components

The class `HS11 Scientific Device`, subclass of both `HS2 Heritage Science Object` and `D8 Digital Device`, represents the tools employed in scientific procedures. These may include spectrometers, microscopes, radiographic systems, or environmental sensors. Devices are further decomposed through `HS12 Device Component`, enabling the modelling of complex configurations (e.g., detectors, stages, light sources), each of which can be linked via `HSP9 used component` to the scientific activity in which it is employed. Devices may also be attributed to their manufacturer (`HSP18 has maker`), supporting provenance and technical attribution.

4.2.2. Digital objects

Digital outputs are central to the production, processing, and dissemination of knowledge in Heritage Science, since they provide the medium through which scientific measurements are preserved, interpreted, and shared. CRMhs includes classes and properties that enable detailed modelling of datasets, software environments, and analysis settings, ensuring semantic traceability and fostering reuse.

4.2.2.1. Scientific datasets

The class `HS13 Scientific Dataset` (subclass of `D9 Data Object`) is used to represent digital resources generated by scientific activities, such as raw data files, processed results, or interpretative outputs. These datasets are often the result of an analysis (`HSP16 produced dataset`) and may in turn serve as input to further investigations (`HSP12 used dataset`).

Datasets may originate from automatic processes or be manually compiled by researchers, and they often consist of a variety of file types (e.g. text, image, numerical, 3D models).

4.2.2.2. Software

The model reuses `D14 Software` to describe digital applications involved in data collection, control of devices, simulation, and processing. Software is linked to activities via `HSP10 used software`, and to datasets via `HSP11 was created using software`, capturing its generative or supportive role in the analytical process. This is crucial to enable reproducibility and to document the digital provenance of results.

4.2.2.3. Analysis settings

Parameters, calibration values, and technical configurations that influence an analysis are represented through the class `HS14 Analysis Settings`. Each setting is associated with an analysis via `HSP13 used settings` and may have specific values recorded with `HSP14 has settings values`. Settings can be directly embedded or specified in an external dataset or file using `HSP15 is specified in`. These elements are fundamental to contextualising measurements and ensuring interpretative reliability, especially when results are reused across projects or studies.

5. Examples of Heritage Science workflows

5.1 Radiocarbon dating of Richard III skeleton

Heritage Science investigations frequently involve complex chains of material transformation, analytical measurement, computational processing and interpretative reasoning. A particularly compelling example of such a workflow is provided by the radiocarbon dating undertaken during the scientific investigation of the skeletal remains discovered in 2012 beneath a municipal car park in Leicester, on the site of the former Greyfriars friary. Archaeological excavation revealed a human skeleton whose location and burial characteristics were consistent with historical descriptions of the interment of Richard III, the last Plantagenet king of England, who died in 1485 at the Battle of Bosworth. The discovery rapidly became the focus of an interdisciplinary research programme combining archaeological investigation, osteological examination, genetic analysis and radiocarbon dating. Within this framework, radiocarbon analysis played an essential role in establishing the chronological consistency of the remains with the historically documented date of the king's death[13].

To obtain an absolute chronological estimate, a small fragment of bone was carefully extracted from the skeleton for laboratory analysis. This intervention was designed to minimise physical impact while ensuring that the material would yield reliable measurements of the isotopic ratio between radiocarbon and stable carbon.

[13] A comprehensive analysis of this case study can be found in Bayliss, A., Bronk Ramsey, C. and van der Plicht, J. 2022. "The skeleton in the car park." In A. Bayliss, C. Bronk Ramsey, J. van der Plicht (eds), *Radiocarbon Dating and Chronological Modelling: Guidelines and Best Practice*, section 5.2. London: Historic England. https://historicengland.org.uk/images-books/publications/radiocarbon-dating-chronological-modelling/.

In the laboratory, the bone sample underwent chemical pretreatment to remove contaminants potentially introduced during burial. The purified material was then analysed using Accelerator Mass Spectrometry (AMS) at two specialised radiocarbon laboratories, the Scottish Universities Environmental Research Centre (SUERC) at the University of Glasgow and the Oxford Radiocarbon Accelerator Unit (ORAU) at the Research Laboratory for Archaeology and the History of Art, University of Oxford. These facilities determined the ratio of ^{14}C to ^{12}C atoms remaining in the sample.
These measurements generated primary numerical datasets expressing the radiocarbon age in years before present together with associated uncertainties. Since radiocarbon measurements do not directly correspond to calendar years, the results were subsequently calibrated against established atmospheric reference curves, producing probability distributions expressed within a calendrical timescale. The final stage of the investigation involved the interpretative synthesis of these calibrated chronological ranges with archaeological context and historical documentation, thereby contributing to the broader identification of the skeletal remains.

5.1.1. Ontological encoding of the scientific workflow

The application of the CRMhs ontology to this radiocarbon investigation allows the entire analytical procedure to be represented as a semantically interconnected knowledge graph. In this framework, each stage of the workflow, from the archaeological discovery of the remains to the generation and interpretation of chronological datasets, is formally represented through the integration of CIDOC CRM, CRMarchaeo, CRMdig and CRMhs classes and properties. This modelling approach ensures that the relationships between material entities, scientific activities, instruments, datasets and interpretative conclusions remain fully transparent and traceable.

5.1.2. Defining the study object and its spatial context

The workflow begins with the archaeological excavation that led to the discovery of the human skeleton at the Greyfriars site in Leicester. This event is formally modelled as an instance of `A9 Archaeological Excavation` from the CRMarchaeo extension, during which the skeletal remains were encountered (`O19 was object encountered through`). Once retrieved and documented, the skeleton is classified as an instance of `E19 Physical Object`. Within the CRMhs framework, `E19 Physical Object` is declared as a subclass of `HS6 Study Object`, meaning that the remains automatically function as the central study object of the Heritage Science investigation while retaining their ontological identity as a physical artefact.

In order to record precisely the location from which material is removed for analysis, a specific portion of the skeleton is identified as an instance of `HS8 Study Object Area`. This area represents the exact surface region selected for the extraction of bone material destined for radiocarbon measurement. The relationship between the `HS6 Study Object` and this selected region is established using the property `HSP17 has area of interest`. The spatial position of this area on the surface of the skeletal element is further documented through `HSP21 has object surface coordinates`, ensuring that the origin of the sampled material can be precisely reconstructed.

5.1.3. Tracking the sampling process and material origin

The removal of the bone fragment intended for radiocarbon analysis constitutes a scientific intervention that transforms part of the original object into a new analytical entity. This event is represented as an instance of `HS1 Scientific Activity`, describing the sampling procedure performed by laboratory specialists. Through this activity, a portion of the skeletal remains becomes an instance of `HS9 Study Object Sample`, a new physical entity created for scientific investigation.

Although this sample exists as an independent entity, it retains an explicit semantic link to its source. The relationship between the `HS9 Study Object Sample` and the parent `HS6 Study Object` is expressed through the property `HSP19 was taken from`, ensuring that the provenance of the sample remains permanently documented. At the same time, the more precise point of origin on the skeleton is preserved through `HSP20 was taken from area`, which connects the sample to the previously defined `HS8 Study Object Area`. Together these links maintain a clear chain of custody that traces the sampled material back to the exact portion of the original object from which it derives.

5.1.4. Sample preparation and chemical transformation

Before measurement can occur, the extracted bone fragment must undergo chemical treatment designed to remove contaminants that could distort the radiocarbon signal. This preparatory phase is modelled as an instance of `HS10 Sample Preparation`. As a specialised form of `HS1 Scientific Activity`, this step records the transformation of the raw bone fragment into a purified laboratory sample suitable for isotopic measurement.

The relationship between this activity and the material it processes is expressed through the property `HSP5 was prepared by`, linking the `HS10 Sample Preparation` event to the `HS9 Study Object Sample`. The analytical procedures applied during this phase are documented through `HSP6 used method` and `HSP7 used protocol`, allowing the specific chemical pretreatment strategy employed by the laboratory to be formally recorded. These procedures typically involve the removal of secondary carbonates and organic contaminants, resulting in a purified collagen fraction ready for measurement.

5.1.5. Isotopic analysis and data production

The determination of the radiocarbon content of the prepared bone sample constitutes the central analytical step of the workflow. This measurement is modelled as an instance of `HS3 Analysis` acting upon the sample through the property `HSP4 was analysed by`. The activity is carried out by the SUERC (or the ORAU) personnel, represented as instances of `E21 Person`, who perform the analysis (`P14 carried out by`) within a specialised laboratory environment represented as an instance of `E53 Place` and linked via the `P7 took place at` property.

The analysis typically consumes the prepared sample as part of the measurement process and generates digital information representing the isotopic composition of the material.

The instrumentation employed in this process is represented as an instance of `HS11 Scientific Device` corresponding to the Accelerator Mass Spectrometry system used to measure the ^{14}C to ^{12}C ratio. The connection between the device and the analytical activity is recorded through the property `HSP8 used device`. The software environment responsible for the calibration and chronological modelling of the radiocarbon measurements is represented as an instance of `D14 Software`. In the present example, we represent the calibration of the measured radiocarbon ages performed using the software *OxCal*, developed at the Oxford Radiocarbon Accelerator Unit, which implements internationally adopted calibration curves such as the *IntCal* series. This software is linked to the analytical workflow through the `HSP10 used software` property.

The numerical results generated during the measurement, including radiocarbon ages expressed in years BP together with isotopic values such as $\delta^{13}C$ and associated uncertainties, are represented as instances of `HS13 Scientific Dataset`, which store values corresponding to radiocarbon ages, isotopic ratios and associated measurement uncertainties. These datasets are explicitly connected to the analytical activity through the property `HSP16 produced dataset`, ensuring that every digital output can be traced back to the scientific process that generated it.

5.1.6. Data calibration and final synthesis

Radiocarbon ages obtained through measurement do not correspond directly to calendar dates and must therefore be calibrated against established atmospheric reference curves. This stage is represented as a subsequent `HS1 Scientific Activity` that operates upon the previously generated `HS13 Scientific Dataset`. The relationship between this activity and the datasets it processes is captured through the properties `HSP2 was performed on` and `HSP12 used dataset`.

Through this calibration process, new chronological values are derived that express the most probable calendar age ranges corresponding to the measured radiocarbon values. These calibrated results are represented as instances of `E54 Dimension` whose values are obtained through the observation mechanism `O16 observed value`. The calibrated chronological information thus emerges as the outcome of a structured sequence of analytical and computational steps rather than a direct measurement.

The final stage of the workflow consists of an interpretative synthesis that relates the calibrated chronological results to the archaeological object from which the sample originated. This interpretative act is modelled through the CIDOC CRM mechanism of `E13 Attribute Assignment`. By employing the properties `P140 assigned attribute to` and `P141 assigned`, the calibrated chronological dimensions are formally associated with the skeletal remains, transforming analytical measurements into historically meaningful knowledge.

Throughout the process, every dataset produced during sampling, preparation, measurement and calibration is represented as an instance of `HS13 Scientific Dataset`. These datasets may be associated with persistent identifiers, versioning information and storage metadata and may be linked to digital repositories or research infrastructures. In this way the entire

radiocarbon workflow, from the discovery of the skeleton to the final chronological interpretation, is represented as a coherent network of entities and activities. The example demonstrates how CRMhs enables the modelling of the physical transformation of material, the succession of scientific operations, the configuration of instruments and software environments, and the interpretative acts through which analytical evidence contributes to the reconstruction of historically significant events.

The whole process is illustrated in *Figure 3*.

This radiocarbon dating example demonstrates how a single analytical objective unfolds into a network of interrelated entities and events and how CRMhs can model physical transformation of material and the succession of scientific activities, methodological choices, instrumental configurations, digital processes and interpretative acts that together produce historical knowledge from archaeological evidence. This ensures that the entire analytical chain remains semantically transparent and reproducible.

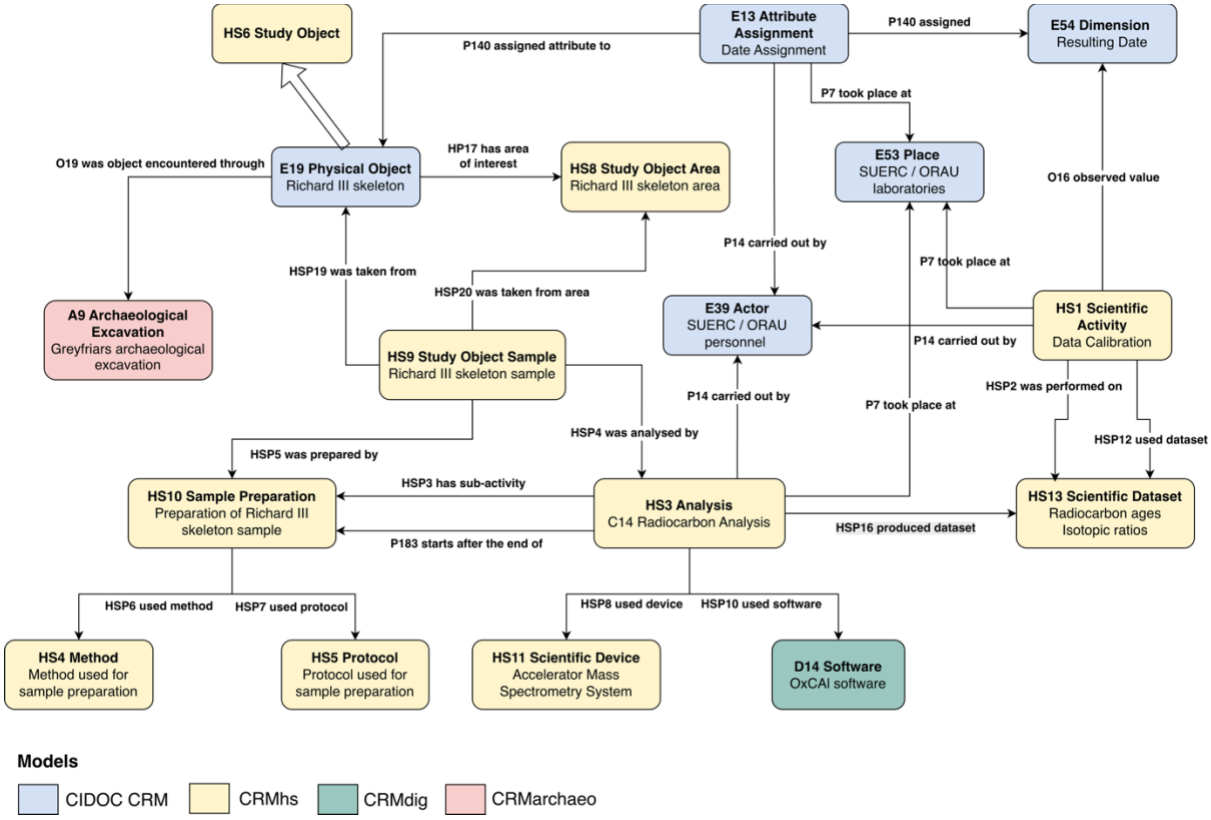

Figure 3. Radiocarbon dating of an archaeological artefact modelled using CRMhs.

Workflows of this type can be established for each of the scenarios defined for Heritage Science combining the various conceptual elements involved in different ways and linking them, according to the needs, by means of specific relationships. The CRMhs model aims to provide the conceptual building blocks to allow the efficient modelling of each scientific activity and build tailored schemas for each type of analysis.

5.2. MA-XRF analysis of Raphael's Leo X portrait

The adoption of high-resolution elemental mapping has transformed the diagnostic approach to masterworks, allowing for a comprehensive study of an artist's palette without any physical contact or damage to the surface. To illustrate how CRMhs handles such non-invasive and large-scale digital investigations, this section explores the macro X-ray fluorescence (MA-XRF) scanning of Raphael's *Portrait of Leo X with two cardinals* painting. Carried out during a significant conservation project at the Opificio delle Pietre Dure, this investigation utilized a portable scanner to meticulously map the distribution of chemical elements across the entire panel, providing insights into the execution phases of the painting before its historical exhibition in Rome[14].

In contrast to the invasive nature of radiocarbon sampling, MA-XRF represents a strictly non-destructive approach that achieves elemental characterization through the induction of secondary X-ray fluorescence. The process relies on a mobile measuring head equipped with a miniature Moxtek X-ray tube, which serves as the primary excitation source. This specific component is engineered to emit a highly stable and collimated beam of X-rays, which, upon striking the painting's surface, causes the atoms of the pigments to emit characteristic fluorescence. By scanning the panel in a systematic grid, the system records localized elemental signatures that are subsequently synthesized into spatial distribution maps. These visual datasets allow experts to identify specific materials, such as vermilion identified through mercury levels, bismuth black via bismuth signals, or the presence of glass powders in red lakes indicated by manganese markers.

The application of the CRMhs ontology to MA-XRF scanning facilitates the integration of physical hardware configurations, digital acquisition parameters, and spatial data, and ensures that the relationship between the scanning device, the specific areas of the painting analysed, and the resulting elemental maps is formally documented and reproducible.

5.2.1. Defining the study object and its spatial context

In terms of data modelling, the *Portrait of Leo X* painting is instantiated as an `E22 Human-Made Object`. Due to the ontological alignment within CRMhs, which recognizes `E22 Human-Made Object` as a subclass of `HS6 Study Object`, the artwork is immediately assigned an investigative role. This ensures its scientific attributes are captured without overriding its primary identification as a physical human-made object in traditional heritage documentation. While the technical goal of the analysis was a total surface mapping, the investigation resulted in an "almost full" coverage. The physical constraints imposed by the painting's weight and the structure of the easel meant that the scanner could not access certain peripheral areas. In the model, this effectively scanned region is documented as a specific instance of `HS8 Study Object Area`.

[14] Mazzinghi, A., Ruberto, C., Giuntini, L., Mandò, P. A., Taccetti, F., & Castelli, L. (2022). Mapping with Macro X-ray Fluorescence Scanning of Raffaello's *Portrait of Leo X*. *Heritage*, *5*(4), 3993-4005. https://doi.org/10.3390/heritage5040205

Because MA-XRF is a mapping technique, the spatial documentation is particularly rigorous. The link between the painting and the scanned area is established through the `HSP17 has area of interest` property. Furthermore, each pixel in the resulting maps is related to the physical surface of the painting through `HSP21 has object surface coordinates`, which provides a 2D spatial referencing framework suitable for close-range documentation.

5.2.2. Technical configuration and analysis settings

The scanning process is modelled as an `HS3 Analysis`, which is a specialized subclass of `HS1 Scientific Activity`. This activity is characterized by its specific technical setup and environmental parameters. The model describes this configuration as follows:

- The `HS11 Scientific Device` class is used to describe the portable MA-XRF scanner developed by the INFN-CHNet.
- Key hardware elements, such as the Moxtek X-ray tube and the Amptek SDD detector, are modelled using the `HS12 Device Component` class and linked via the `HSP9 used component` property.
- Specific operational parameters are captured through the `HS14 Analysis Settings` class. These include the 30 kV anode voltage, 90 µA filament current, and the scanning velocity of 10 mm/s used during the analysis and directly linked to it via the `HSP13 used settings` property.

5.2.3. Data acquisition and digital outputs

The scanning activity generates a set of digital resources that represent the distribution of chemical elements across the panel. Each of these results is modelled as an `HS13 Scientific Dataset` and is connected to the `HS3 Analysis` through the `HSP16 produced dataset` property.

These datasets include elemental distribution maps for lead (Pb), mercury (Hg), copper (Cu), iron (Fe), and manganese (Mn). For instance, the detection of bismuth (Bi) in the Pope's cassock and the metallic detail of the handbell provides evidence for the use of "bismuth black," a rare pigment characteristic of Raphael’s Roman period. Similarly, the `HS13 Scientific Dataset` for manganese (Mn) suggests the presence of glass powder in the red lakes, likely used as a drying agent for the oil medium.

5.2.4. Interpretative synthesis and final results

The final phase involves the cross-referencing of these elemental maps with other diagnostic data, such as IR reflectography or stratigraphic cross-sections. This interpretive process is modelled via the `I7 Belief Adoption` class of CRMinf. Through this synthesis, the researchers could confirm that the painting was conceived as a triple portrait from the beginning (`I7 Belief`) as the copper (Cu) distribution in the background is uniform and does not show evidence of a different initial composition. This transformation of numeric data and elemental images into historical and technical knowledge is documented using the `E13 Attribute Assignment` mechanism, used to directly associate the acquired information to

the painting, ensuring that the entire analytical chain, from the configuration of the X-ray tube to the identification of the artist's technique, remains semantically transparent.

Figure 4 below offers a detailed overview of the whole process

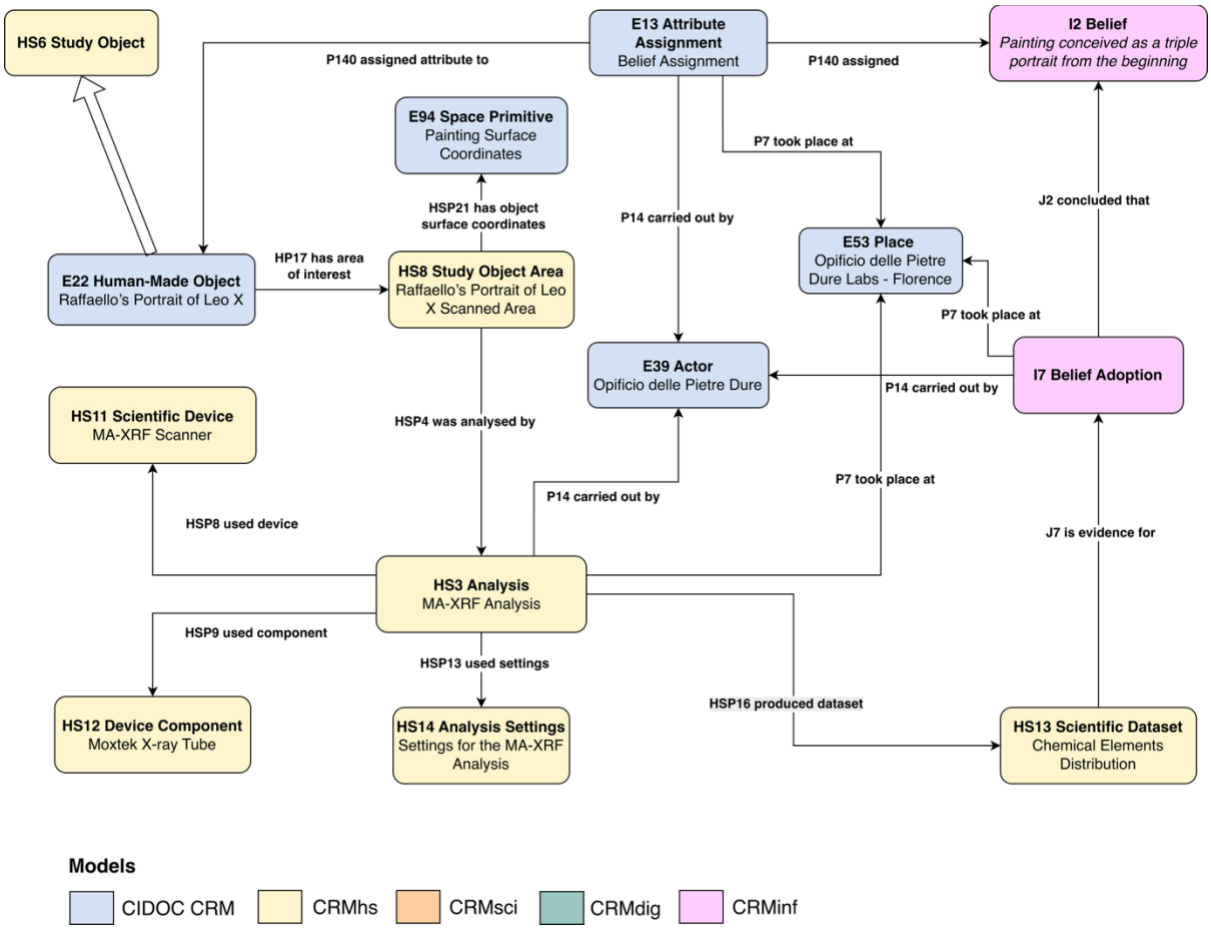

Figure 4. CRMhs encoding of Raffaello's Portrait of Leo X MA-XRF analysis.

6. CRMhs Classes and properties

6.1. CRMhs Class Hierarchy

E1 CRM Entity
E2 - - Temporal Entity
E3 - - - Condition State
I2 - - - Belief
E4 - - - Period
E5 - - - - Event
E7 - - - - - Activity
HS1 - - - - - Scientific Activity
HS3 - - - - - - Analysis
HS10 - - - - - - Sample Preparation
E13 - - - - - - Attribute Assignment
S4 - - - - - - - Observation
HS3 - - - - - - - Analysis
E11 - - - - - Modification
HS10 - - - - - - Sample Preparation
I1 - - - - - Argumentation
I7 - - - - - - Belief Adoption
E77 - Persistent Item
E70 - - Thing
E71 - - - Human-Made Thing
E24 - - - Physical Human-Made Thing
E22 - - - - Human-Made Object
HS12 - - - - - Device Component
D8 - - - - - Digital Device
HS11 - - - - - - Scientific Device
E28 - - - - Conceptual Object
E55 - - - - - Type
E90 - - - - - Symbol Object
E73 - - - - - - Information Object
D1 - - - - - - - Digital Object
D9 - - - - - - - - Data Object
HS13 - - - - - - - - - Scientific Dataset
D14 - - - - - - - - Software
E29 - - - - - - - Design or Procedure
HS4 - - - - - - - Method
HS5 - - - - - - - Protocol
HS14 - - - - - - - Analisys Settings
E41 - - - - - - Appellation
E35 - - - - - - Title
E94 - - - - - - Space Primitive
HS2 - - Heritage Science Object
HS6 - - - Study Object

E18 - - - - Physical Thing
HS7 - - - - Study Object Portion
HS8 - - - - Study Object Area
HS9 - - - - Study Object Sample
HS12 - - - Device Component
S10- - Material Substantial
S11- - - Amount of Matter
S13- - - - Sample
HS9 - - - - Study Object Sample
E39 - - Actor
E74 - - - Group
E21 - - - Person
E70 - - Thing
E52 - Time-Span
E53 - Place
HS8 - Study Object Area
E54 Dimension

6.2. CRMhs Properties List

Property	Entity Domain	Entity Range
HSP1 has activity title	HS1 Scientific Activity	E35 Title
HSP2 was performed on	HS1 Scientific Activity	HS2 Heritage Science Object
HSP3 has sub-activity	HS1 Scientific Activity	HS1 Scientific Activity
HSP4 analysed	HS3 Analysis	HS6 Study Object
HSP5 prepared	HS10 Sample Preparation	HS9 Study Object Sample
HSP6 used method	HS1 Scientific Activity	HS4 Method
HSP7 used protocol	HS1 Scientific Activity	HS5 Protocol
HSP8 used device	HS1 Scientific Activity	HS11 Scientific Device
HSP9 used component	HS1 Scientific Activity	HS12 Device Component
HSP10 used software	HS1 Scientific Activity	D14 Software
HSP11 was created using software	HS13 Scientific Dataset	D14 Software
HSP12 used dataset	HS1 Scientific Activity	HS13 Scientific Dataset
HSP13 used settings	HS3 Analysis	HS14 Analysis Settings
HSP14 has settings values	HS14 Analysis Settings	E54 Dimension
HSP15 is specified in	HS14 Analysis Settings	HS13 Scientific Dataset
HSP16 produced dataset	HS1 Scientific Activity	HS13 Scientific Dataset
HSP17 has area of interest	HS6 Study Object	HS8 Study Object Area
HSP18 is located within	HS7 Study Object Portion	HS8 Study Object Area
HSP19 was taken from	HS9 Study Object Sample	HS6 Study Object

HSP20 was taken from area	HS9 Study Object Sample	HS8 Study Object Area
HSP21 has object surface coordinates	HS8 Study Object Area	E94 Space Primitive
HSP22 has maker	HS11 Scientific Device	E39 Actor
HSP23 has model	HS11 Scientific Device	E55 Type

6.1 CRMhs Classes Declarations

The classes of the CRMhs model are comprehensively declared in this section using the following format:

- Class names are presented as the title of the section describing them. CRMhs classes are all named following a single schema: the string "HS" is used as a prefix, followed by an identification number, and a name that indicates the type of resource rendered by the class.
- The line "Subclass of:" declares the superclass of the class from which it inherits properties.
- The line "Superclass of:" is a cross-reference to the subclasses of this class.
- The line "Scope note:" contains the textual definition of the concept the class represents.
- The line "Examples:" contains a bulleted list of examples of instances of this class.
- The line "Properties:" declares the list of the class's properties. Each property is represented by its unique identifier, its forward name and its inverse name in parentheses.
- Inherited properties are not represented.
- Properties of properties, if they exist, are provided indented and in parentheses beneath their respective domain property.

HS1 Scientific Activity

Subclass of: `E7 Activity`

Superclass of: `HS3 Analysis`, `HS10 Sample Preparation`

Scope note: This class comprises all the actions that are performed in the course of a scientific investigation, with the purpose of acquiring, validating, or structuring knowledge about a given object, sample, or phenomenon. It includes a broad spectrum of operations that occur in research contexts, such as the organisation and execution of analytical workflows, the preparation of physical materials for measurement, the configuration and operation of devices, the processing of data, and the use of software tools for computation or visualisation. An instance of `HS1 Scientific Activity` is usually bound to a specific time-span and presupposes the participation of agents (human or institutional), the use of techniques, tools, or protocols, and the targeting of specific materials or digital entities. This class provides the overarching event layer that connects abstract procedures (such as methods and protocols) to their actual execution in a reproducible context. It serves as a generalisation for specialised types of activity represented in the CRMhs model, notably `HS3`

Analysis (for scientific measurements) and HS10 Sample Preparation (for pre-analytical transformations).

Examples:

- The interpretative synthesis (HS1) performed to relate the calibrated chronological ranges with the archaeological context and historical documentation of Richard III (see section 5.1.6).

Properties:

- HSP1 has activity title (is activity title of)
- HSP2 was performed on (was subject of)
- HSP3 has sub-activity (is sub-activity of)
- HSP6 used method (was method used by)
- HSP7 used protocol (was protocol used by)
- HSP8 used device (was device used by)
- HSP9 used component (was component used by)
- HSP10 used software (was software used by)
- HSP12 used dataset (was dataset used by)
- HSP16 produced dataset (was dataset produced by)

HS2 Heritage Science Object

Subclass of: E70 Thing

Superclass of: HS6 Study Object, HS11 Scientific Device, HS12 Device Component, HS13 Scientific Dataset

Scope note: This class comprises all entities, whether physical or digital, that can be involved in a scientific investigation either as objects of study or as devices, carriers, or outputs of scientific processes. It includes individual artefacts, materials, and specimens, as well as software, digital files, and datasets. An instance of HS2 Heritage Science Object represents a persistent and identifiable item that maintains its identity over time and across investigative contexts, regardless of whether it undergoes physical or conceptual transformation. This class functions as a foundational node for more specific types of entities defined in this model, such as HS6 Study Object (entities intentionally selected as the focus of research), HS9 Study Object Sample (physically extracted portions), HS11 Scientific Device (tools or devices), and HS13 Scientific Dataset (structured outputs of analytical activities).

HS3 Analysis

Subclass of: HS1 Scientific Activity, S4 Observation

Scope note: This class comprises scientific activities that are carried out to detect, measure, or determine specific physical, chemical, or qualitative properties of an object, sample, or environmental phenomenon. Analyses typically involve the application of established methods and protocols, the use of devices and software, and the production of structured data. Examples include spectrometric techniques (such as XRF, FTIR, or Raman), chromatographic analyses (such as GC-MS or HPLC), as well as non-invasive imaging or sensor-based observations. Both destructive and non-destructive forms of analysis are included. An instance of HS3 Analysis is characterised by a defined temporal span, the use of well-documented procedures, and the production of results

that can be interpreted in relation to a study object or its components. It frequently appears as part of a larger scientific workflow, often following a sample preparation phase (HS10), and results in the creation of one or more scientific datasets (HS13). This class serves to explicitly model the moment when empirical knowledge is generated through the controlled application of techniques to observable entities.

Examples:

- The radiocarbon analysis (HS3) performed on the one sample of the Richard III skeleton (HS9) to determine the $^{14}C/^{12}C$ ratio (see section 5.1.5).
- The MA-XRF analysis (HS3) of Raphael's 'Portrait of Leo X' (HS6) carried out by the team of the Opificio delle Pietre Dure in Florence to produce high-resolution elemental distribution maps (see section 5.2.2).

Properties:

- HSP4 analysed (was analysed by)
- HSP13 used settings (settings were used for)

HS4 Method

Subclass of: E29 Design or Procedure

Scope note: This class comprises abstract, generalised descriptions of technical or scientific procedures that can be repeatedly applied to achieve a specific type of investigative or analytical result. A method defines a standardised set of conceptual steps or operations, often documented in literature or through expert practice, that are independent of a particular time, place, or execution. Examples of methods include radiocarbon dating, Raman spectroscopy, principal component analysis, and finite element simulation. An instance of HS4 Method provides a reusable semantic anchor to relate different scientific activities that share the same underlying conceptual process, regardless of variations in device, software, or implementation details. This abstraction enables knowledge models to distinguish between *what kind of technique* is being applied and *how exactly* it is carried out in a given case. By modelling methods explicitly, this class supports the traceability of scientific reasoning and fosters comparability across studies that adopt similar or compatible procedures.

Examples:

- The chemical pretreatment method (HS4) used by the ORAU laboratory technicians to remove potential contaminants from the bone fragment of Richard III skeleton (see section 5.1.4).

HS5 Protocol

Subclass of: E29 Design or Procedure

Scope note: This class comprises formalised and prescriptive plans that define how a scientific method (HS4) is to be operationalised in a specific investigative context. A protocol describes, with a high level of detail, the sequence of actions to be performed, the required devices and materials, the parameters

to be set, and the conditions under which the procedure is to be considered valid. Protocols are often documented as standard operating procedures, laboratory handbooks, or workflow templates, and they serve to ensure consistency, repeatability, and reproducibility in scientific practice. Unlike methods, which are conceptual and general, protocols are context-bound and provide concrete instructions that can be followed during a Scientific Activity (HS1), especially in processes such as Sample Preparation (HS10) or Analysis (HS3) . An instance of HS5 Protocol may refer to one or more methods and is typically associated with specific laboratory environments, device, or datasets. It enables the alignment of scientific workflows with established good practices and quality control standards.

Examples:

- The standard operating procedure (HS5) followed by the ORAU laboratory technicians to ensure the correct execution of the chemical purification on the bone sample of Richard III skeleton (see section 5.1.4).

HS6 Study Object

Subclass of: HS2 Heritage Science Object

Superclass of: HS7 Study Object Portion, HS8 Study Object Area, HS9 Study Object Sample, E18 Physical Thing

Scope note: This class comprises physical objects or artefacts that have been intentionally selected as the primary subjects of a scientific investigation. The HS6 class does not define a specific material typology, but rather groups heterogeneous physical entities defined in CIDOC CRM according to their epistemic status, namely their being intentionally selected and subjected to scientific investigation through instances of HS1 Scientific Activity. In this sense, HS6 Study Object operates as a role-based aggregation comparable to E72 Legal Object, which brings together diverse entities on the basis of their being subject to rights. By being declared as a superclass of E18 Physical Thing, HS6 explicitly anchors the model within the material domain of CIDOC CRM while preserving conceptual coherence with the distinction between natural and human-made entities. This alignment ensures that all physical objects involved in Heritage Science investigations, whether naturally occurring materials such as sediments or bones, or culturally produced artefacts and structures, can be uniformly integrated within the same upper-level conceptual framework. At the same time, this choice reinforces semantic interoperability with the broader CIDOC CRM ecosystem, allowing HS2 instances to inherit well-established properties and constraints while remaining open to the inclusion of digital and computational entities that extend beyond the strictly physical domain. Therefore, HS6 may include archaeological finds, human remains, natural specimens, and other organic or inorganic materials, as well as artworks and artefacts such as paintings, frescoes, statues, reliefs, and architectural elements. A Study Object is defined by its epistemic role: it is not merely present in the scientific context, but is actively interrogated through observation, measurement, or experimentation with the goal of generating or refining knowledge. It may be investigated as a whole, or through the isolation of portions (HS7 Study Object Portion) or

spatially defined areas (HS8 Study Object Area). It is of particular importance to maintain accurate records of the origin and provenance of Study Objects, including the conditions under which they were removed from their original contexts, the methods used for their preservation, and any interventions carried out prior to analysis. This is essential to relate analytical results to their historical, cultural, and material contexts in a scientifically valid and ethically responsible way. Many Study Objects are accompanied by existing metadata produced by museums, archives, or other heritage institutions. In some cases, they may already be described using CIDOC CRM-compliant documentation and can be referenced via persistent identifiers.

Examples:

- The skeleton of Richard III (E19 subclass of HS6) encountered by the archaeologists during the Greyfriars archaeological excavation and subsequently selected for absolute chronological determination (see section 5.1.2).
- Raphael's 'Portrait of Leo X' (E22 subclass of HS6), an oil on panel painting selected by the researchers of the Opificio delle Pietre Dure for a high-resolution MA-XRF diagnostic campaign (see section 5.2.1).

Properties:

- HSP17 has area of interest (is area of interest of)

HS7 Study Object Portion

Subclass of: HS6 Study Object

Scope note: This class comprises physically bounded parts of a Study Object (HS6) that are identified and delimited for the purpose of scientific investigation, without necessarily being removed or transformed into separate material samples. Typical examples include stratigraphic layers on a wall painting, fragments of a ceramic vessel that remain in situ, or individual anatomical elements within a human skeleton. A Study Object Portion is materially continuous with its parent object and retains its structural, spatial, and contextual relationship to the whole. It is distinguished from a Study Object Sample (HS9), which involves an act of extraction and implies a new instance with a distinct material trajectory. The identification of such portions may be based on visual criteria (e.g. cracks, pigment differentiation), functional or anatomical logic, or other scientific rationales. Their documentation supports the localisation and interpretation of analytical results in relation to the original physical entity. Study Object Portions are especially useful in Heritage Science contexts where invasive sampling is undesirable or prohibited, yet fine-grained spatial or material discrimination is required.

Example:

- The individual anatomical element within a skeleton (HS7) selected by the researchers for morphological study without physical detachment from the find (see section 4.2.1.3).

Properties:

- HSP18 is located within (contains portion)

HS8 Study Object Area

Subclass of: `HS6 Study Object`, `E53 Place`

Scope note: This class comprises spatial areas of artefacts or other physical objects that are occupied by the portions of material selected for scientific analysis. These areas are understood as instances of places situated on or within the geometry of a Study Object (`HS6`), and can be identified using relative or absolute coordinates, grid systems, or spatial reference frames. A Study Object Area provides a spatial anchor for linking observations, sampling actions, or analytical results to a specific location on the object, without necessarily implying a separable material unit. It is particularly relevant in cases where scientific investigations are localised but non-invasive, or where results need to be visualised or correlated spatially across a surface or volume. This class enables precise documentation of the spatial context in which a portion (`HS7`) or a sample (`HS9`) is situated, and supports reproducibility, comparison, and spatial reasoning in scientific workflows. It is essential for research domains that rely on in situ analysis, imaging, mapping, or micro-spatial interpretation.

Example:

- The effectively scanned region of Raphael's 'Portrait of Leo X' (`HS8`) documented by the researchers of the Opificio delle Pietre Dure to identify the actual surface analysed versus the total geometry of the painting (see section 5.2.1).

Properties:

- `HSP21 has object surface coordinates (object surface coordinates of)`

HS9 Study Object Sample

Subclass of: `HS6 Study Object`, `S13 Sample`

Scope note: This class comprises portions of material that have been physically extracted from a Study Object (`HS6`) for the purpose of scientific investigation within a heritage context. An instance of `HS9 Study Object Sample` is a newly individuated physical entity resulting from an act of separation, and is subsequently processed, prepared, or analysed as part of a scientific workflow. As a subclass of `S13 Sample` (CRMsci), this class inherits the general characteristics of scientific samples. However, it is specifically intended to represent samples derived from cultural heritage objects and is characterised by a constitutive and enduring relationship with the object of provenance. Unlike a Study Object Portion (`HS7`), which remains in situ, a Study Object Sample implies physical detachment and potential modification of its original material context. However, a Study Object Sample is not merely detached material, it remains semantically and historically linked to the cultural entity from which it originates and continues to be documented, managed, and preserved within the heritage domain. This continuing relationship implies that the sample may inherit aspects of the identity, significance, custodial responsibility, and conservation framework associated with the originating Study Object. Examples include micro fragments of paint layers, mortar samples extracted from historic masonry structures, parchment fibres taken

from ancient manuscripts, wood samples collected from sculpted objects for dendrochronological examination . An instance of HS9 Study Object Sample may retain structured information concerning its provenance, extraction conditions, and related areas (HS8) of the originating object. The identification, storage, and tracking of such samples are essential for ensuring traceability, accountability, and continuity of documentation across analytical procedures. Samples may undergo further transformation through Sample Preparation activities (HS10) and are often altered, partially consumed, or destroyed during analysis.

Examples:

- The bone sample(HS9) physically extracted from the skeleton of Richard III to be used for absolute chronological determination (see section 5.1.3).
- A micro-fragment of a paint layer (HS9) detached by the researchers from a historic masonry structure to undergo chemical characterization operations (see section 4.2.1.2).

Properties:

- HSP19 was taken from (was object source of sample)
- HSP20 was taken from area (was area source of sample)

HS10 Sample Preparation

Subclass of: HS1 Scientific Activity, E11 Modification

Scope note: This class comprises scientific activities performed to prepare a material Heritage Science Sample (HS9) for subsequent analysis. These activities are applied exclusively to samples that have already been physically extracted from their source and are intended to bring the material into a condition suitable for analysis, measurement, observation, or device processing. Typical operations include cleaning, drying, grinding, polishing, filtering, embedding in resin, or mounting on supports. Each procedure aims to enhance the accessibility, stability, or compatibility of the sample with specific analytical techniques, while minimising contamination or degradation. Sample Preparation is typically guided by established Protocols (HS5) and may involve the use of specialised devices (HS11) and components (HS12). It typically precedes the execution of an Analysis (HS3) and plays a critical role in ensuring the quality, reproducibility, and interpretability of scientific results.

Examples:

- The chemical purification (HS10) of Richard III skeleton's bone sample performed by the ORAU laboratory technicians to remove potential contaminants that could bias the radiocarbon results (see section 5.1.4).
- The chemical pretreatment procedures (HS10) carried out by the researchers to transform the raw wood fragment into a laboratory-ready state for AMS measurement (see section 5.1.4).

Properties:

- HSP5 prepared (was prepared by)

HS11 Scientific Device

Subclass of: HS2 Heritage Science Object, D8 Digital Device

Scope note: This class comprises devices specifically designed or configured for use in scientific investigations, particularly in the context of Heritage Science. Scientific devices are man-made physical artefacts that perform essential functions in activities such as Sample Preparation (HS10) and Analysis (HS3). They include both general-purpose laboratory devices and highly specialised analytical equipment such as spectrometers, chromatographs, electron microscopes, and imaging systems. Each device operates as a coherent unit and may integrate various functional subsystems (HS12 Device Component). Scientific devices are typically described in protocols (HS5), and their identification, configuration, and usage are crucial for ensuring the reproducibility and comparability of results. The documentation of which device was used, and under what conditions, also supports validation and interpretative transparency across datasets. This class excludes purely conceptual or virtual tools (e.g. software libraries or algorithms), which are modelled separately within the broader investigative workflow.

Examples:

- The Accelerator Mass Spectrometry (AMS) system (HS11) utilized by the researchers at the ORAU laboratory to measure the isotopic ratio of the prepared bone samples extracted from Richard III skeleton (see section 5.1.5).
- The portable MA-XRF scanner (HS11) developed by the INFN-CHNet and employed by the team of the Opificio delle Pietre Dure during the diagnostic campaign on Raphael's painting (see section 5.2.2).

Properties:

- HSP22 has maker (made)
- HSP23 has model (is model of)

HS12 Device Component

Subclass of: HS2 Heritage Science Object, E22 Human-Made Object

Scope note: This class comprises identifiable physical components that form part of a Scientific Device (HS11) and contribute to its overall functioning. A Device component represents a functionally meaningful part of a larger device, such as a sensor, radiation source, optical lens, detector, stage controller, or other technical element, which may be replaced, configured, or described independently. Device components are not used in isolation, but only in conjunction with a parent device. Their presence, configuration, and calibration may significantly affect the outcome of scientific activities, particularly in Sample Preparation (HS10) and Analysis (HS3). Documenting device components enables the tracking of specific hardware setups, the reproducibility of experimental conditions, and the comparison of results obtained under different configurations. It also supports maintenance, versioning, and quality assurance procedures in scientific infrastructures. This class does not include conceptual or virtual submodules (e.g. firmware, software plugins), which are documented separately.

Examples:

- The Moxtek X-ray tube (HS12) utilized by the researchers of the Opificio delle Pietre Dure as the primary excitation source to produce a stable and precisely collimated X-ray beam (see section 5.2.2).
- The Amptek SDD detector (HS12) integrated into the portable MA-XRF scanner by the INFN-CHNet team to record localized elemental signatures during the diagnostic campaign (see section 5.2.2).

HS13 Scientific Dataset

Subclass of: HS2 Heritage Science Object, D9 Data Object

Scope note: This class comprises structured sets of scientific data produced, transformed, or reused in the course of a scientific investigation. A Scientific Dataset may result from measurement processes (HS3 Analysis), be generated during computational workflows, or serve as an input to subsequent research activities. It can consist of raw sensor output, calibrated values, processed matrices, or interpretative datasets derived from analytical results. Scientific datasets may take the form of digital files, database tables, spreadsheets, image stacks, or multidimensional arrays. They are often associated with a specific Scientific Activity (HS1), and may reflect the use of particular software tools, protocols, or devices. An instance of HS13 Scientific Dataset is understood as a meaningful and self-contained information object, documented for the purpose of traceability, reuse, and interpretation. Its structure, format, language, and provenance should be explicitly recorded.

Example:

- The raw isotopic counts and associated uncertainties (HS13) produced by the researchers during the AMS measurement of the bone sample taken from Richard III skeleton (see section 5.1.5).
- The elemental distribution maps for lead (Pb), mercury (Hg), and copper (Cu) (HS13) generated by the researchers of the Opificio delle Pietre Dure during the MA-XRF scanning of Raphael's painting (see section 5.2.3).

Properties:

- HSP11 was created using software (was software used to create)

HS14 Analysis Settings

Subclass of: E29 Design or Procedure

Scope note: This class comprises the settings and controlled parameters that define the specific conditions under which a scientific analysis or measurement was performed. These may include environmental factors (e.g., temperature, humidity), device configurations (e.g., voltage, acquisition time), or any other experimental variables influencing the analysis. Settings can be documented to ensure reproducibility and provide context for interpreting results. Instances of this class are typically associated with a specific instance of HS3 Analysis.

Examples:

- The anode voltage of 30 kV (HS14) configured by the researchers for the MA-XRF scanning of Raphael's painting (see section 5.2.2).
- The filament current of 90 µA (HS14) set by the researchers during the technical setup of the MA-XRF analysis (see section 5.2.2).

- The scanning velocity of 10 mm/s (HS14) established by the researchers to define the operational parameters of the portable scanner (see section 5.2.2).

6.2. CRMhs Properties

This section presents in detail all the properties of the CRMhs model. A simple notation is used for description. In particular, property axioms are stated using the following common template:

- Property names are presented as the title of the section describing them. CRMhs properties are all named following a single schema: the string "HSP" is used as a prefix, followed by an identification number, and a name that indicates the type of relationship captured by the property. The name of the inverse property is also reported in parentheses.
- The line "Domain:" declares the class for which the property is defined.
- The line "Range:" declares the class to which the property points, or that provides the values for the property.
- The line "Subproperty of:" is a cross-reference to any superproperties the property may have.
- The line "Superproperty of:" is a cross-reference to any subproperties the property may have.
- The line "Scope note:" contains the textual definition of the concept the property represents.
- The line "Examples:" contains a bulleted list of examples of instances of this property.

HSP1 has activity title (is activity title of)

Domain: `HS1 Scientific Activity`

Range: `E35 Title`

Scope Note: This property associates a scientific activity with a title used to identify or describe it. The title may derive from laboratory documentation, project records, published reports, or internal workflows, and is typically assigned to facilitate reference, classification, or citation. Unlike an appellation, which may cover alternative or common names, the title usually reflects a formal or conventional designation of the scientific activity.

Examples:

- The MA-XRF scanning of Raphael's painting (`HS1`) has activity title (`HSP1`) "Macro X-ray fluorescence scanning of Raphael's Portrait of Leo X" (`E35`) as designated by the researchers of the Opificio delle Pietre Dure (see section 5.2).

HSP2 was performed on (was subject of)

Domain: `HS1 Scientific Activity`

Range: `HS2 Heritage Science Object`

Scope Note: This property links a Scientific Activity to the Heritage Science Object it was carried out on. It refers to any material entity involved as the subject of the operation, whether a study object, a portion, a sample, a scientific device, or a component. It is used to specify the material focus of operations such as

preparation, analysis, calibration, maintenance, manipulation, or disposal, and enables the traceability of scientific procedures and results to their physical or technical source.

Examples:

- The data calibration activity (`HS1`) was performed on (`HSP2`) the raw radiocarbon dataset (`HS13`) by the ORAU laboratory personnel to obtain calendar dates (see section 5.1.6).

HSP3 has sub-activity (is sub-activity of)

Domain: `HS1 Scientific Activity`

Range: `HS1 Scientific Activity`

Scope Note: This property links a scientific activity to another activity that was carried out as part of it. It is used to represent complex procedures composed of multiple, sequential or parallel operations, such as a sample preparation followed by multiple analyses. Sub-activities inherit the temporal, procedural, and contextual framework of the parent activity, but may have their own specific targets, devices, or protocols. This property enables the modelling of structured workflows and supports the decomposition of scientific procedures into meaningful units of action.

Example:

- The radiocarbon dating workflow (`HS1`) has sub-activity (`HSP3`) the chemical purification of the wood fragment (`HS10`) (see section 5.1.4 and *Figure 3).*

HSP4 analysed (was analysed by)

Domain: `HS3 Analysis`

Range: `HS6 Study Object`

Scope Note: This property links an analysis activity to the Heritage Science object it aimed to characterise. It is used to specify that the object, portion, or sample was the subject of a scientific measurement or observation. It differs from was performed on by indicating that the purpose of the activity was specifically analytical, i.e. to detect, quantify, or assess the properties of the object.

Examples:

- The MA-XRF analysis (`HS3`) analysed (`HSP4`) Raphael's 'Portrait of Leo X' (`HS6`) to map the distribution of chemical elements across the panel surface (see section 5.2.2).
- The isotopic analysis (`HS3`) performed by the researchers analysed (`HSP4`) the bone sample of Richard III skeleton (`HS9`) to determine the $^{14}C/^{12}C$ ratio (see section 5.1.5).

HSP5 prepared (was prepared by)

Domain: `HS10 Sample Preparation`

Range: `HS9 Study Object Sample`

Scope Note: This property links a sample preparation activity to the sample it was applied to. It indicates that the activity transformed or conditioned the sample—typically after its extraction—to make it suitable for a specific type of analysis. Preparation may involve physical, chemical, or mechanical operations such as

cleaning, drying, grinding, embedding, or mounting. The prepared sample may subsequently undergo further processing, alteration, or even destruction during later scientific procedures.

Examples:

- The chemical purification process (`HS10`) prepared (`HSP5`) the bone fragment from Richard III skeleton (`HS9`) for the subsequent AMS measurement (see section 5.1.4).

HSP6 used method (was method used by)

Domain: `HS1 Scientific Activity`

Range: `HS4 Method`

Scope Note: This property links a scientific activity to the method it followed. The method provides a general, abstract description of the conceptual procedure underlying the activity, independently of its specific execution. It allows different activities to be related to the same methodological approach, supporting traceability, reproducibility, and semantic comparability across investigations.

Examples:

- The chemical pretreatment of the bone sample taken from Richard III skeleton (`HS10`) used as method (`HSP6`) the chemical purification to remove potential contaminants (see section 5.1.4).

HSP7 used protocol (was protocol used by)

Domain: `HS1 Scientific Activity`

Range: `HS5 Protocol`

Scope Note: This property links a scientific activity to the protocol that guided its execution. It identifies the prescriptive plan that defined the sequence of steps, conditions, and resources to be used. The protocol reflects the procedural choices adopted in a given context and ensures the reproducibility of the activity.

Examples:

- The chemical pretreatment of the bone sample taken from Richard III skeleton (`HS10`) used as protocol (`HSP6`) the standard laboratory protocol to remove potential contaminants (see section 5.1.4).

HSP8 used device (was device used by)

Domain: `HS1 Scientific Activity`

Range: `HS11 Scientific Device`

Scope Note: This property links a scientific activity to the device employed during its execution. It refers to the physical device used to perform, support, or enable the procedure, including devices for measurement, manipulation, sample conditioning, or imaging. The relation expresses the functional involvement of the device in the activity, regardless of whether it operated alone or as part of a more complex configuration.

Examples:

- The MA-XRF analysis (HS3) used device (HSP8) the portable MA-XRF scanner developed by the INFN-CHNet (see section 5.2.2).

HSP9 used component (was component used by)

Domain: HS1 Scientific Activity
Range: HS12 Device Component
Scope Note: This property links a scientific activity to a component of a device that was involved in its execution. It refers to any identifiable physical part—such as a sensor, source, lens, or stage controller—that contributed to the functioning of the setup. The relation highlights the role of specific components within the activity, allowing for the documentation of configurations, replacements, or calibrations relevant to the procedure.
Examples:

- The MA-XRF analysis (HS3) used component (HSP9) the Moxtek X-ray tube as the primary excitation source (see section 5.2.2).

HSP10 used software (was software used by)

Domain: HS1 Scientific Activity
Range: D14 Software
Scope Note: This property links a scientific activity to the software used during its execution. It includes computational tools employed for data acquisition, control of devices, processing, simulation, or visualisation. The relation documents the functional involvement of software in the activity and enables the traceability of digital environments that shaped its execution.
Examples:

- The isotopic analysis of the bone sample taken from Richard III skeleton (HS3) used software (HSP10) the acquisition software to record and manage the raw isotopic counts (see section 5.1.5).

HSP11 was created using software (was software used to create)

Domain: HS13 Scientific Dataset
Range: D14 Software
Scope Note: This property links a scientific dataset to the software used in its generation. It applies when the dataset results from computational processes such as data processing, simulation, modelling, or reconstruction. The relation identifies the software as a generative tool and contributes to documenting the digital provenance of the dataset. This property is a shortcut of the more fully developed path from HS13 Scientific Dataset through P94 was created by, E65 Creation, P6 used specific object, D14 Software.

HSP12 used dataset (was dataset used by)

Domain: HS1 Scientific Activity
Range: HS13 Scientific Dataset
Scope Note: This property links a scientific activity to a dataset it made use of. The dataset may have served as a source of input values, reference information, calibration

parameters, or any other form of structured data necessary to execute the activity. This relation establishes a dependency between the activity and pre-existing digital content.

Examples:

- The interpretative data calibration activity (`HS1`) performed by the ORAU researchers to validate the archaeological sequence used dataset (`HSP12`) containing the probability distributions to transform analytical measurements derived from the analysis of the bone sample taken from Richard III skeleton into verified historical knowledge (see section 5.1.6).

HSP13 used settings (settings were used for)

Domain: `HS3 Analysis`

Range: `HS14 Analysis Settings`

Scope Note: This property links a scientific analysis to the analysis settings applied during its execution. It captures the parameters, configurations, or conditions that influenced the acquisition of data.

Examples:

- The MA-XRF analysis (`HS3`) used setting (`HSP13`) the anode voltage of 30 kV (see section 5.2.2).
- The scanning session (`HS3`) used setting (`HSP13`) the scanning velocity of 10 mm/s (see section 5.2.2).

HSP14 has settings values (are settings values of)

Domain: `HS14 Analysis Settings`

Range: `E54 Dimension`

Scope Note: This property specifies the numerical value associated with a given analysis setting. It enables the documentation of measurable parameters relevant to the scientific analysis. Specific values can be represented through the CIDOC CRM class `E54 Dimension` and further detailed using properties `P90 has value` (for the numeric content) and `P91 has unit` (for the measurement unit). These values, together with their contextual meaning, contribute to a more accurate interpretation and reproducibility of the analysis.

Example:

- The anode voltage setting (`HS14`) has setting value (`HSP14`) 30 kV (`E54`) as specified in the instrument configuration log for the MA-XRF scanning of Raphael's 'Portrait of Leo X' (see section 5.2.2).

HSP15 is specified in (contains specification of)

Domain: `HS14 Analysis Settings`

Range: `HS13 Scientific Dataset`

Scope Note: This property links a set of analysis settings to the digital file or dataset in which they are specified. Such files may include configuration files, device setup logs, or any structured data source that defines the operational parameters applied during the analysis.

Example:

- The analytical settings for the macro X-ray fluorescence scanning of Raphael's 'Portrait of Leo X' (`HS14`) are specified in (`HSP15`) the technical log file *LeoX_OPD_XRF_v1.log* (`HS13`) used during the investigation at the Opificio delle Pietre Dure. (see Section 5.2).

HSP16 produced dataset (was dataset produced by)

Domain: `HS1 Scientific Activity`
Range: `HS13 Scientific Dataset`
Scope Note: This property links a scientific activity to a dataset that resulted from it. It indicates that the dataset captures the structured output of the activity, such as raw measurements, processed data, or interpreted results. The relation supports the documentation of data provenance and the traceability of scientific outputs.
Examples:

- The AMS analysis of the bone sample taken from Richard III skeleton (`HS3`) produced dataset (`HSP16`) the raw isotopic counts and associated uncertainties (see section 5.1.5).
- The scanning session (`HS3`) produced dataset (`HSP16`) the elemental distribution maps for lead and mercury (see section 5.2.3).

HSP17 has area of interest (is area of interest of)

Domain: `HS6 Study Object`
Range: `HS8 Study Object Area`
Scope Note: This property links a study object to the area of interest on its surface that is relevant for a given scientific investigation. It serves to highlight the spatial region of the object that has been selected for observation, measurement, or sampling. The area may correspond to a visible feature, a predefined section used in repeated experiments, or a region determined by the configuration of the scientific devices used. This explicit connection supports the localisation of scientific operations, facilitates spatial comparisons, and enhances reproducibility by allowing subsequent studies to focus on the same part of the object.
Examples:

- Raphael's 'Portrait of Leo X' (`HS6`) has area of interest (`HSP17`) the effectively scanned region of the panel (see section 5.2.1).
- The skeleton of Richard III (`HS6`) has area of interest (`HSP17`) the specific spot identified by the researchers for sampling (see section 5.1.2).

HSP18 is located within (contains portion)

Domain: `HS7 Study Object Portion`
Range: `HS8 Study Object Area`
Scope Note: This property links a Study Object Portion (`HS7`) to the Study Object Area (`HS8`) within which it is located. It expresses a spatial inclusion relationship between a physically identifiable portion and a broader, documented area of the same study object. The relation is useful to contextualise measurements or interventions made on a specific portion with respect to a defined spatial

reference, supporting spatial reasoning, repeatability, and visual documentation of the investigation context.

Example:

- The blue *lapis lazuli* inlay of the Mask of Tutankhamun (`HS7`) is located within (`HSP18`) the spatial area of the headcloth (`HS8`), providing a precise link between the mineral component and its position on the golden mask of the pharaoh (Egyptian Museum, Cairo).

HSP19 was taken from (was object source of sample)

Domain: `HS9 Study Object Sample`

Range: `HS6 Study Object`

Scope Note: This property links a sample to the study object from which it was physically extracted. It expresses a material derivation and implies that the sample originated from a specific object, portion, or area selected for investigation. The relation is essential for tracking the origin of sampled material and maintaining its connection to the broader context of study.

Examples:

- The bone sample (`HS9`) was taken from (`HSP19`) the skeleton of Richard III (`HS6`) and used as sample for a dating analysis (see section 5.1.3).

HSP20 was taken from area (was area source of sample)

Domain: `HS9 Study Object Sample`

Range: `HS8 Study Object Area`

Scope Note: This property associates a sample with the area of the study object from which it was physically extracted. It enables documentation of the precise spatial origin of a sample when a more general link to the whole object or a portion of it (`HSP18 was taken from`) is insufficient. This level of detail is particularly relevant when spatial heterogeneity within the object influences the results of the analysis, or when the same object yields multiple samples from distinct areas.

Examples:

- The bone sample (`HS9`) was taken from a designed area on the skeleton of Richard III (`HSP20`) corresponding to the specific spot on the skeleton itself (`HS8`) (see section 5.1.3).

HSP21 has object surface coordinates (object surface coordinates of)

Domain: `HS8 Study Object Area`

Range: `E94 Space Primitive`

Scope Note: This property associates a study object area with a set of spatial coordinates that describe its position and extent on the surface of a study object. These coordinates do not represent geographical or abstract spatial positions, but rather the location of the area directly on the physical object. They can be expressed using 2D image-based systems (e.g., pixel regions), relative dimensions (e.g., millimetres from a reference point), or other spatial referencing frameworks suitable for close-range documentation. The property is essential for precisely identifying the analysed area, especially in non-

invasive techniques such as MA-XRF or multispectral imaging, where the region of interest must be defined without altering the object.

Examples:

- The scanned region of Raphael's 'Portrait of Leo X' (`HS8`) has object surface coordinates (`HSP21`) the 2D spatial referencing framework mapping the pixel regions to the physical surface of the panel, providing a high-resolution grid for close-range documentation (see section 5.2.1).

HSP22 has maker (made)

Domain: `HS11 Scientific Device`

Range: `E39 Actor`

Scope Note: This property links a scientific device, including any of its identified components, to the agent responsible for its creation. It may refer to an individual, institution, or manufacturer who designed, assembled, or produced the device or one of its constituent parts. The relation enables documentation of technical provenance and attribution across all levels of scientific hardware. This property is a shortcut of the more fully developed path from `HS11 Scientific Device` through `P108 was produced by`, `E12 Production`, `P14 carried out by`, `E39 Actor`.

Examples:

- The portable MA-XRF scanner (`HS11`) *has maker* (`HSP22`) the INFN-CHNet research infrastructure (see section 5.2.2).

HSP23 has model (is model of)

Domain: `HS11 Scientific Device`

Range: `E55 Type`

Scope Note: This property links a scientific device to its specific technical or commercial model. It allows the documentation of the series, specification, or product identifier under which the device was manufactured or is commonly known. This can include references to the manufacturer's catalogue name, a version code, or other formal designations that distinguish one configuration or generation of device from another.

Examples:

- The large-area MA-XRF scanner (`HS11`) utilized at Rijksmuseum Amsterdam for the comprehensive mapping of Rembrandt's 'Night Watch' has model (`HSP23`) "Bruker M6 JETSTREAM" (`E55`).

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